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Abstract

The **BlastSmart App** project focuses on developing a tool that assists with blast design and fragmentation calculations for surface mining operations. The application consists of two main components: the **Surface Blasting Design Calculator** and the **Fragmentation Calculator**. The aim of the project is to simplify blast design and improve the accuracy of fragmentation predictions.

The application uses engineering parameters such as blast hole diameter, bench height, explosive properties, and rock type to perform the required calculations. As part of the project, algorithms and pseudocode were developed for both calculators before they were implemented in Python.

By automating these calculations, the BlastSmart App reduces manual errors, saves time, and assists mining engineers in making informed decisions during blast planning. The project demonstrates how software can be used to improve the efficiency, accuracy, and safety of surface mining operations.

1 Background

Surface blasting is widely used in mining for rock excavation. Effective blast design relies on precise calculations of parameters such as hole diameter and explosive density, which can be time-consuming and error-prone when performed manually. The BlastSmart App automates these calculations through two integrated tools: the **Surface Blasting Design Calculator** and the **Fragmentation Calculator**. These tools utilize engineering formulas to provide fast and accurate results, reducing the likelihood of human error. By combining mining engineering principles with computer programming, the BlastSmart App improves blast planning, enhances decision-making, and contributes to safer and more efficient surface mining operations.

2 Introduction

Blasting is a method used in surface mining to break rock into smaller fragments using explosives. The effectiveness of a blast depends on selecting suitable blast design parameters such as bench height, hole diameter, explosive density, powder factor, and spacing-to-burden ratio. If these parameters are selected incorrectly, problems such as poor fragmentation, excessive flyrock, and increased operational costs may occur.

The **BlastSmart App** was developed to automate blast design and fragmentation calculations. By replacing manual calculations with computer software, the application improves accuracy, saves time, and assists mining engineers in designing effective blasts and predicting rock fragmentation.

3 Objectives

3.1 Main Objective

The main objective of this project is to develop the **BlastSmart App**, which automates surface blasting design and fragmentation calculations for surface mining operations.

3.2 Specific Objectives

- To study the principles of surface blasting and rock fragmentation.
- To identify the key blast design parameters required for the BlastSmart App.
- To develop algorithms and pseudocode for both the **Surface Blasting Design Calculator** and the **Fragmentation Calculator**.
- To implement both calculators using Python.
- To test the calculators using the appropriate engineering formulas.
- To demonstrate how automation can improve blast planning, accuracy, and efficiency in mining operations.

4 Literature Review

4.1 Surface Blasting

Surface blasting is a rock fragmentation method utilizing explosives. It entails drilling holes, inserting explosives, and detonating them systematically. Effective blast design enhances productivity, cost-efficiency, and safety by minimizing risks such as rock flying and ground vibration.

4.2 Blast Design Parameters

Factors influencing blast planning include the height of the rock face, drill hole diameter, explosive density, explosive quantity, spacing of holes, and plug length. These parameters impact the efficacy of explosives and the resulting fragmentation of rocks.

4.3 Rock Fragmentation

Rock fragmentation refers to the breakage of rock into smaller pieces post-blasting, facilitating loading, hauling, and crushing processes, thereby conserving energy. The effectiveness of the fragmentation is influenced by rock type, explosive selection, and blast design.

4.4 BlastSmart Application

The BlastSmart App features two tools: the Surface Blasting Design Calculator and the Fragmentation Calculator. It employs engineering formulas for blast calculations, enhancing accuracy, minimizing errors, and expediting blast planning.

4.5 Importance of Digital Tools in Mining

Digital tools enable mining engineers to perform calculations swiftly and precisely. The BlastSmart App automates these calculations, aiding decision-making and improving safety, productivity, and efficiency in mining operations.

5 Results and Discussion

The Blast Smart App is designed to provide on-site testing and validation of blast parameters for optimal fragmentation, significantly reducing errors compared to manual calculations. Good fragmentation lowers crushing and secondary breaking costs, and the App allows users to input specific parameters, facilitating an efficient trial-and-error approach before actual blasting. This ensures manageable fragmentation for cleaning crews, enabling the loading of fragmented rock without requiring additional rebreaking.

Controllable parameters include burden, spacing, stemming length, and explosive energy, while uncontrollable parameters include geological and geotechnical characteristics. The App plays a crucial role in dynamically adjusting controllable parameters based on drilled information, thereby ensuring proper drilling geometry for effective fragmentation. Fragmentation directly influences mucking productivity, crusher energy consumption, plant efficiency, yield, and recovery.

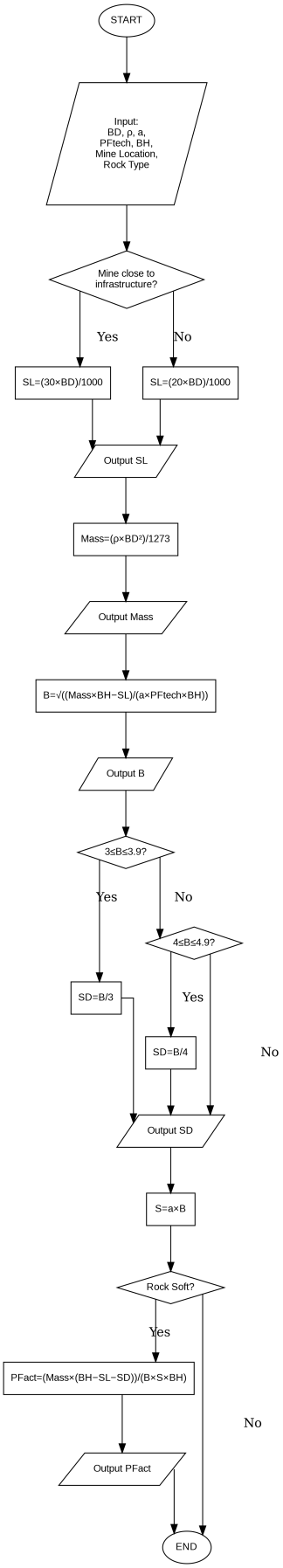
The implemented algorithm automates calculations for stemming length, burden, power factor, and mean fragment size using user-provided inputs. In addition, the application provides predictive capabilities that allow blast designs to be adjusted according to production targets and operational requirements.

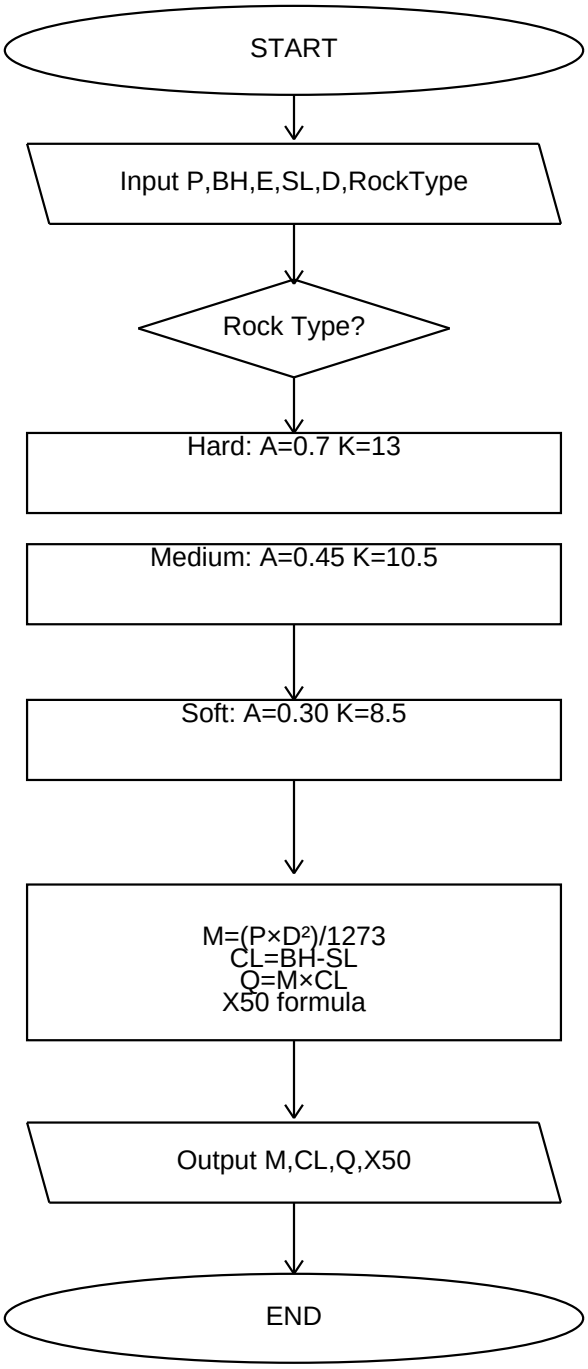
6 Conclusion and Future Work

The implementation of the Blast Smart App represents a significant advancement in the optimization of mining production processes. It establishes a digital blasting tool foundation that improves workflow efficiency and enables engineers to predict blasting outcomes more effectively.

Future development should focus on integrating the application with drilling operations and geological databases, allowing analyses to incorporate additional inputs from geologists and drill operators, such as hole diameter and rock mass characteristics. Further enhancements may also include modules for calculating Peak Particle Velocity (PPV) and predicting dust generation resulting from blasting operations.

Surface Blasting Design Flowchart




START
Input P,BH,E,SL,D,RockType
Rock Type?
Hard: A=0.7 K=13
Medium: A=0.45 K=10.5
Soft: A=0.30 K=8.5
M=(P×D²)/1273
CL=BH-SL
Q=M×CL
X50 formula
Output M,CL,Q,X50
END